Checkpoint 2: Counting Events

> **!** Important
>
> **Due 11/15** (12 points). See Project Central for the full project overview. Make sure you have addressed any feedback from Checkpoint 1.

Goal

Model a **count** in your scenario with a discrete random variable, and use it to answer your guiding question: a planning question for your public health program, such as how many follow-up appointments to reserve, how much support to plan for, or how often an unusual year happens.

Course topics used: discrete random variables, PMFs and CDFs, expected value and variance, named discrete distributions (Bernoulli, Binomial, Poisson).

Your scenario

Use the Checkpoint 2 rows from your Checkpoint 0 table, with any corrections from my feedback.

Start by copying your **Checkpoint 2 guiding question** from below (or Checkpoint 0 if you made your own) into the top of your editing file, with any changes from my feedback. Everything in this checkpoint builds toward answering it: you will calculate its probability in Task 3, check it with a simulation in Task 4, and answer it in one sentence in Task 6. The example for each scenario is below. If you changed the cutoff in Checkpoint 0 (for example, 8 appointments instead of 6), use your own version.

Scenario A

Community breast cancer screening program

- **Random variable:** X = number of screens read as abnormal out of the 40 screens on one van day.
- **Distribution:** Binomial, with $n = 40$ (course value) and p = the abnormal interpretation rate from Lee et al. (2023).
- **Guiding question (example from Checkpoint 0):** If the program reserves 6 follow-up appointments after each van day of 40 screens, what is the probability that it needs more than 6?

Scenario B

County COVID-19 testing and contact tracing

- **Random variable:** X = number of the 3 household contacts of one case who get infected.
- **Distribution:** Binomial, with $n = 3$ (course value) and p = the household secondary attack rate for Omicron from Madewell et al. (2022).
- **Guiding question (example from Checkpoint 0):** For a case who lives with 3 other people, what is the probability that at least 2 of them also get infected?

Scenario C

State newborn heart screening program

- **Random variable:** X = number of babies born with CCHD in the state in one year.
- **Distribution:** Binomial, with $n = 40{,}000$ births (course value) and p = the prevalence of CCHD from the Cochrane review. Because n is large and p is small, you may also use a Poisson distribution with $\lambda = np$. If you do, explain why the Poisson is a good approximation here, using the rule of thumb on the Lesson 12 slide Poisson Approximation of the Binomial. (Check carefully which part of the rule of thumb your n and p meet!)
- **Guiding question (example from Checkpoint 0):** With 40,000 births a year, what is the probability that the state has 30 or more babies with CCHD in one year?

Tasks

1. Define your random variable

- Write X in words, including its units and time frame, using the definition for your scenario above (can use wording above).

- List the possible values (the support) of X.

2. Justify the distribution

- State the distribution for your scenario and its parameter(s), citing where each value came from.
- List the assumptions of that distribution (e.g., a fixed number of trials, independent trials, and the same probability of a "yes" on every trial), and explain whether each is reasonable for your scenario.
- Identify the assumption **most likely to fail** in real life, and explain why.

3. Solve it analytically

- Find $E(X)$, $\text{Var}(X)$, and $\text{SD}(X)$ using formulas from class.
- Calculate the probability in your **Checkpoint 2 guiding question**.
  - Show the formula (PMF or CDF), then compute with R (`dbinom()`, `pbinom()`, `dpois()`, or `ppois()`, from Lessons 7, 9, and 12).

4. Simulate it

- Simulate at least 100,000 values of X in R.
- Make a plot comparing the **simulated** distribution with the **theoretical PMF**.
- Compare the simulated mean, variance, and guiding-question probability with the analytic values in a small table.

> Where to start
>
> We've seen these in class! Copy the code from these slides into your editing file and adapt it to your scenario:
>
> - **Simulate and summarize:** Lesson 12, Ghost problem revisited: simulate it. Replace the two parameters with your n and p, and rename `Y` to `X`. Add your guiding-question probability to the `summarise()`. For example, `mean(X > 6)` is the proportion of simulations with $X > 6$, which estimates $P(X > 6)$. To compare, put your analytic values next to the simulated ones.
> - **Theoretical PMF table:** Lesson 7, Falls in Older Adults (4/8). This makes a tibble with every value of x and its probability from `dbinom()`.
> - **Plot:** Lesson 7, Falls in Older Adults (7/8). The bars are your simulated values, and the red dots are the theoretical PMF. When the dots sit on top of the bars, your simulation matches the PMF.

Tip for Scenario C: plot only the values that happen

With $n = 40{,}000$, copying `x = 0:n` from Lesson 7 makes a table with 40,001 rows, and `scale_x_continuous(breaks = 0:n)` tries to draw 40,001 tick marks! Almost all of those values have a probability of essentially 0. Instead:

- Use `x = 0:60` in your PMF table. This covers all the values that come up in practice.
- Use `scale_x_continuous(breaks = seq(from = 0, to = 60, by = 5))` for a tick mark every 5 babies.

If you use the Poisson distribution, use `rpois()` and `dpois()` in place of `rbinom()` and `dbinom()`, with `lambda` in place of `size` and `prob`.

5. Stretch (optional, +1 bonus)

Break the assumption you identified in Task 2 and show how the answer changes. Some ideas for each scenario:

- **Scenario A:** The abnormal rate may differ from one van day to the next (for example, more women getting their first-ever mammogram on some days). Let p vary from day to day and compare $P(X > 6)$ with your original answer.
- **Scenario B:** Household members are not independent: if one contact gets infected, the others are more likely to be exposed too. Simulate households where p itself varies from household to household, and compare $P(X \geq 2)$ with your original answer.
- **Scenario C:** The number of births and the prevalence may change from year to year. Let n or p vary across years and compare the chance of a very busy year.

Hint: letting p vary

In your simulation, add a column that draws a different p for each simulation, then use that column in `rbinom()`. For example, `runif()` (Lesson 8) draws a value evenly between two numbers:

```
mutate(p_today = runif(n(), min = 0.05, max = 0.12),
       X = rbinom(n(), size = 40, prob = p_today))
```

Pick a range centered on your original p, so the average rate stays about the same. Then only the variability changes.

6. Say it

Write a short paragraph (4–6 sentences) **for the program manager** who needs to plan around this count. Use plain language, with no probability notation. Answer these questions in order. Each one uses something you already found:

Question for the program manager	Use this from your work	Sentence starter
1. What should I expect on a typical day, household, or year?	$E(X)$ and SD(X) from Task 3	"On a typical _____, expect about _____, give or take about _____."
2. How often does the situation in our guiding question happen?	Your guiding-question probability from Task 3 (and your simulation in Task 4), written as a natural frequency with a probability in parentheses.	"This happens about _____ out of every _____ days (households, years)."
3. What could make these numbers wrong?	The assumption most likely to fail from Task 2, and whether it would make the situation in Question 2 more or less common (use your Stretch result if you did it)	"Our model assumes _____. If that is not true, _____ will happen more (or less) often."
4. So what should we plan for?	Your answers to Questions 1–3	"We recommend _____."

> Tip: turning a probability into a natural frequency
>
> Multiply your probability by a round number of days, households, or years. For example, a probability of 0.045 is about 4.5 out of 100, or about 1 out of every 22 (because $1/0.045 \approx 22$). Pick whichever is easiest for the program manager to picture.

Then answer your **Checkpoint 2 guiding question** in one sentence, using the probability (for example, "The probability that _____ is about _____.").

Submission

Submit a rendered HTML file (use `code-fold: true`) and your `.qmd`. Make sure the qmd and html are named `lastname_firstinitial_checkpoint_02.qmd` and `lastname_firstinitial_checkpoint_02.html` or `lastname1_lastname2_checkpoint_02.qmd`

and `lastname1_lastname2_checkpoint_02.html`. One submission per team. Either person in the pair can submit it.

You can download the `.qmd` editing file for this checkpoint here. The link takes you to GitHub, where you can click the download button (the arrow pointing down) near the top right of the file.

The editing file has the headings and tasks, but not the detailed instructions on this page. **Please do not remove anything from the editing file.** You will only add your work to it.